

David Fernando Esparza Esparza

+52 (449) 108-9418 | david.fernando.esp@gmail.com | Aguascalientes, Mexico
davidesperza.github.io | github.com/davidesperza

Professional summary

Senior Engineer with over 7 years of experience in full-stack architecture, system modernization, and data automation. Expert in architecting scalable solutions using Python, SQL, and AWS, with a technical focus on Data Engineering, ETL pipelines, and Machine Learning. Proven track record of optimizing distributed system performance, automating data-heavy workflows, and driving technical leadership in agile environments.

Technical skills

- **Databases & Search:** SQL, MongoDB, Elasticsearch, MeiliSearch.
- **Data & Backend:** Python (Django, Pandas), C# (.NET), Kafka, ETL Pipelines, Core Web Vitals.
- **Cloud, DevOps & Tools:** AWS (S3, EC2), Kubernetes (Lens), Docker, Git, RESTful APIs.
- **Front-End:** React, Astro, JavaScript (ES6+), styled-jsx, Ant Design, UI/UX.
- **Languages:** Spanish (Native), English (Advanced).

Professional experience

Independent Data Engineering Projects | Jan 2026 – Present

Data & Automation Developer

- **Automated ETL Pipelines:** Engineered a data automation system using Python and Pandas to extract academic datasets, transforming raw variables into standardized protocol documents.

iad México | Mexico City, Mexico | July 2021 – September 2025

Full Stack Developer

- **Front-End Modernization:** Led the redesign of the core portal using React and Astro; utilized user analytics and runtime performance data to maximize rendering speed and responsiveness.
- **Containerization & Orchestration:** Managed high-availability Python/Django systems, leveraging Kubernetes and Lens to monitor cluster workloads, logs, and pod resource allocation on AWS infrastructure.
- **Event-Driven Streams:** Integrated Apache Kafka for microservices communication, ensuring reliable, decoupled data streams across corporate applications.
- **Distributed Search:** Optimized system performance and query latency by architecture-level restructuring of Elasticsearch and MeiliSearch clusters.

BOZ | Aguascalientes, Ags. | September 2019 – June 2020

Software Developer

- **System Migration & Integrity:** Executed the complete migration of legacy corporate systems to modern architectures; redesigned relational SQL structures and RESTful APIs to maximize data retrieval efficiency.

Global Hitss | Aguascalientes, Ags. | February 2019 – June 2019

Software Developer

Technical Collaboration: Developed a high-throughput ticketing web portal using C# .NET, coordinating within an agile team of 6 developers to deliver scalable enterprise features.

Education

Polytechnic University of Aguascalientes (2015 - 2019)

B.S. in Strategic Information Systems Engineering